

Mario Hernandez

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EDUCATION

California State University, San Marcos

Aug. 2021-May 2026

Bachelor of Science in Electrical Engineering

SKILLS

Hardware/PCB: Board Bring up, Schematic Capture, Component Selection, PCB Layout, Power Electronics, BOM Generation, Design Verification

Embedded Systems: STM32, GPIO, ADC, PWM, SPI, I²C, UART, SWD debugging, Interrupts, Timers

RF/Wireless: BLE, Impedance Matching, Antenna Basics, S-parameters

Programming & Tools: C/C++, MATLAB, Cadence Virtuoso, LTspice, HFSS

Debugging & Test: Oscilloscope, Multimeter, Power Supply, VNA, Spectrum Analyzer, Logic Analyzer, Root Cause Analysis

WORK EXPERIENCE

Operations Manager- *Queens Restaurant inc.*

Oct. 2022- Present

- Managed daily operations for a high-volume restaurant, overseeing staff scheduling, training, inventory management, and customer service
- Led and trained employees on standard operating procedures, food safety requirements, and quality-control processes
- Coordinated with vendors and suppliers to maintain inventory levels and resolve delivery, quality, and logistics issues
- Implemented process improvements that increased operational efficiency and maintained product consistency during peak-volume periods
- Troubleshooting equipment, workflow, and staffing issues to minimize downtime and maintain business operations
- Managed cash handling, labor allocation, and daily operational performance while ensuring compliance with company standards

PROJECTS

Hardware Design Lead- *Dual DC-DC Converter PCB Design*

May 2026

- Designed and simulated a custom dual-converter PCB providing 12V→5V buck and 5V→12V boost power conversion.
- Selected switching regulators, inductors, capacitors, and feedback networks; developed production-style schematics, PCB layout, BOMs, and ERC/DRC-verified designs in Altium Designer
- Validated converter performance through LTspice analysis of startup behavior, switching waveforms, output regulation, and inductor ripple current.

Hardware Engineering Lead- *Intra-Oral BLE Pressure Sensing Hardware Platform*

Jan. 2025- Nov. 2025

- Designed a wearable Bluetooth-based pressure sensing system to measure tongue force distribution for assistive communication and medical applications
- Designed and implemented a 2-layer flexible PCB integrating 7 FSR sensors, 5V→3.3V regulation, and FPC-based programming for a wearable sensing system
- Developed analog front-end using voltage dividers to interface FSR outputs with the nRF52840 ADC for real-time data acquisition
- Diagnosed unstable sensor readings caused by high-impedance noise ($>1\text{ M}\Omega$) using oscilloscope analysis; implemented RC filtering and optimized resistor values ($1\text{ M}\Omega \rightarrow 500\text{ k}\Omega$) to improve signal stability
- Identified voltage drop across FPC power delivery; redesigned architecture to transmit 5V and regulate locally, restoring reliable operation
- Performed power rail verification, current monitoring, and signal integrity testing during hardware bring-up
- Designed STM32-based programming module for SWD programming and debugging of the nRF52840

Embedded Systems Engineer – *PIR Motion Sensor Alarm System*

Mar. 2024

- Integrated a PIR motion sensor using interrupt-driven GPIO inputs on an STM32 Nucleo-64 board for real-time motion detection
- Developed PWM-based audio control to drive an 8-ohm speaker and generate alternating alarm tones
- Implemented LED indicators and pushbutton interrupt handling for system activation and deactivation
- Performed staged hardware bring-up and signal validation using an oscilloscope and multimeter to verify GPIO timing and PWM waveform integrity